Discrete Event Simulation of a Generic Hospital Emergency Room: An Academic Study of Implications of Pandemic Related Resource Management

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Abstract

A discrete simulation of Pre-Pandemic vs. a Pandemic hospital emergency room resulted in identifying the significant impact that adding new processes, like COVID testing, can have on the efficiency of the emergency room operation. Additional resources would be needed to balance the impact to the test time on the patients processing time and time in the emergency room system. If an emergency room operates on priority by severity level, the patients that will experience the greatest impact to service are the less severe cases. It is also important to note that the new process can be made too efficient and then have negative impacts to queues downstream. Furthermore, with a finite amount of available beds, balancing the number available for general vs. quarantined beds should be driven by data. A knee jerk reaction of designating too many beds for the new disease might make waiting times for day- to-day emergencies difficult to service. Further research should be done to balance this based on positivity rates in the areas.

Introduction – Motivation

Since the start of the COVID-19 pandemic, the healthcare industry has faced unprecedented challenges. One of the most pertinent consequences has been the ongoing stress on the financial stability of hospitals and health systems, which continues to threaten their ability to provide essential services to patients and communities (Milligan & Johnson, 2021). Hospitals around the country have experienced declines in non-COVID-19 admissions, while emergency rooms and ICUs have been overrun with patients suffering from the extreme symptoms of COVID-19 (Birkmeyer et al., 2020). Most importantly, hospitals have reported that the COVID-19 pandemic has significantly strained health care delivery (Grimm, 2021).
It is obvious that hospitals and health systems have been strained to the brink, and they require relief and assistance to meet the health care needs of their patients and communities. Hospitals have had difficulty balancing the complex and resource-intensive care needed for COVID-19 patients with efforts to resume routine hospital care (Grimm, 2021). The emergency department is an area where access to care is vitally important, especially for patients with a lower socioeconomic status. Though, as mentioned above, emergency departments have been negatively impacted by the COVID-19 pandemic in a significant way. So, it is imperative that hospitals ensure that all patients arriving in their emergency departments, especially those with acute medical illnesses, can obtain the proper care as needed during the pandemic to avoid adverse outcomes.

Problem Definition

The purpose of this project was to academically simulate a generic hospital emergency room system and investigate the implications of pandemic scenarios. Our main goal was to compare two models with one another in which each one represented the pre-pandemic and pandemic realities, respectively. We analyzed the wait times for resources (specifically, a bed in the emergency room), the wait times to see a doctor or nurse practitioner for the first time, and the total time patients spent in the system (as well as other patient statistics) for a base model and another that accounted for the changes a hospital might have implemented due to pandemic related consequences. Then, we compared the two systems via the differences in sample means of the listed performance metrics to see if they were statistically distinct from each other.

Along with the comparison of the two simulated systems, a stretch goal was to identify any potential problems within the pandemic adjusted model, such as bottlenecks. One significant complication was identified, so we recommended and tested an enhancement to help alleviate the obstacle, improve the system flow, and decrease patient wait times and time spent in the system.

Literature Survey

First, Jacobson et al. (2006), Hamrock et al. (2013), and Zhang (2018) focused exclusively on how discrete-event simulation has been utilized in health care. Jacobson et al. (2006) explained how to apply discrete-event simulation of health care systems. It also provided a good outline of various simulation studies for different areas of a hospital, including the emergency room, and it talked about the important performance measures associated with health care systems. Hamrock et al. (2013) summarized how to implement a successful simulation study by

describing applicable scenarios and the steps required for development. Plus, it assessed an original model as a practical example. Zhang (2018) conducted a systematic survey of the literature that has been published over the last two decades, and other health care related topics were discussed, such as health and care systems operation.

Next, there were a few papers our group reviewed that were independent research studies. Cubukcuoglu et al. (2020) built a simulation to determine the optimal configuration of hospital space planning while satisfying the required hospital standards. While our project focused on answering an entirely different set of questions, the paper was useful because the assumptions were explicitly stated and discussed, including the probability distributions and parameters that were utilized. Williams et al. (2020) used discrete-event simulation to determine the optimal number of critical care beds required in the ICU and compared different "what-if" scenarios and their outcomes. The paper was incredibly beneficial because it outlined the entire model design, and it provided an in-depth "model validation" table that compared the simulated output with real-world data. Still, our group primarily examined wait times, queue lengths, and patient flow in the emergency room, as opposed to managing bed capacity in the ICU. Hajirizi and Berisha (2019) was the paper most closely related to ours since the authors' goal was to help reduce emergency room wait times in Kosovo hospitals. The process flow diagram of the final simulation model served as a good foundation for our own work, along with the underlying data used to build the model. Above all, these three papers were helpful in building our final baseline simulation model and determining the proper probability distributions and parameters.

Moreover, our group assessed research papers devoted to reducing the impact of COVID-19 on hospitals using simulation modeling. In fact, Currie et al. (2020) discussed how simulation modeling can help support administrators and leaders make the most informed decisions possible. The authors also outlined the different areas of support that are available from the simulation community. For example, Garci-Vicuna et al. (2020) used simulation to estimate the transitory state of the health system in Navarre, Spain and support decision-making concerned with the short-term planning of hospital beds. A flowchart was provided that emphasized the hospitalization needs of COVID-19 patients. A comparable diagram was presented in Melman et al. (2021) that illustrated elective and emergency surgery patient flows during the COVID-19 pandemic. To help alleviate the substantial burden the pandemic has had on scarce resources, the authors utilized a discrete-event simulation model to evaluate the resource allocation strategies for two different hospital scenarios. Furthermore, Lay et al. (2020) focused on the bed occupancy ratio, which is an important management figure to track to proactively avoid overcrowding and maintain a high quality of care. The authors used discrete-event simulation to evaluate the impact COVID-19 has had on a hospital's bed occupancy ratio.

In summary, the literature about COVID-19 were great examples of how our group's baseline simulation model could be adjusted to reflect some of the typical modifications hospitals have made in response to the pandemic. However, the goal of our group's project was notably distinct. Unlike the aforementioned papers, we were concerned with evaluating how the pandemic has impacted emergency room wait times, queue lengths, and patient length of stay by using discrete-event simulation to compare a pre-pandemic baseline model with one altered to account for pandemic related changes.

Proposed Method – Intuition and Description of Approaches

Due to the ongoing problems and restrictions being caused by the pandemic, collecting data from a local hospital emergency room was unfeasible. Instead, the main hospital model we constructed was an amalgamation of the various assumptions, constraints, distributions, etc. that were employed by some of the projects we reviewed in the "Literature Survey."

First, to better compare the two systems, we utilized the "common random numbers" variance reduction technique. So, we used the same patient arrival and service times when simulating the different configurations of the hospital emergency room, which induced positive correlation between them. By subjecting the alternative systems to identical experimental conditions, we hoped to achieve a savings in variance of the differences between the sample means. As a result, we could be much more confident about our declaration if the two systems did present evidence of being different.

Next, each run or replication of the simulation lasted 90 days, and there were a total of 10 replications with 10 different seed numbers for the simulation. Note that Hajirizi and Berisha (2019) used the same figures, and their emergency department process flow served as the main underpinning for our final base model. Furthermore, multiple replications were required to get reasonable half-widths for the mean estimators of the key performance metrics and their corresponding differences.

For patient arrivals, we assumed that the interarrival times followed a Poisson process and were exponentially distributed with a mean of 5 minutes, which was the same assumption made by Cubukcuoglu et al. (2020). In fact, Hamrock et al. (2013), Williams et al. (2020), and Dehghani et al. (2017) also utilized or recommended an exponential distribution to model patient interarrival times. A mean of 5 minutes resulted in a range of 47,300 to 51,500 patients entering the emergency over a 90-day period, which is about 189,200 to 206,000 patients annually. These figures were quite similar to the number of emergency room visits Grady Hospital in Atlanta, GA estimates it handles annually, which is more than 150,000 (Grady Health, 2021). So, we can assume that our generic hospital served a city of about 500,000 people, which was Atlanta's population at the time of the 2020 Census. Finally, we did not cap our waiting room, turn patients away, or account for patients leaving the system due to the waiting times being too long with less than 1% of entities remaining in the system at the conclusion of the simulations.

Since real-world data was difficult to acquire, we created a simulated hospital emergency room based on academic and commercial studies and metrics. Once a patient was generated, we assigned the entity its various service times, which are all considered below. Then, we duplicated (or "separated") the entity and sent one copy to the base model and the other to the pandemic model. This process corresponded to the "common random numbers" technique. Patients arrived at the emergency room department admissions station to be processed, and they were assigned a condition upon their arrival. We assumed that the admissions station resource had a capacity of two staffers and a service time distribution of Lognormal(5,2) (Dehghani et al., 2017). The condition of the patient corresponded to the triage score a patient was expected to be given at the triage station, and these scores are directly related to the 5-level Emergency Severity Index (ESI) used in over half of US emergency departments (Reiter & Scaletta, 2019). Further information about the 5-level system can be found in Table 1.

After completing the admissions process, the patients were routed to the triage station,

which took 5 minutes, where a triage nurse collected relevant information and realistically assigned a priority based on the ESI level of the patient. Please note that aside from the admissions process, the queue types for all subsequent resources were “lowest attribute value” for the ESI level. This meant that the most severe patients were processed first when a resource finally became available. Continuing on, we assumed that the triage block consisted of two stations and nurses (Dehghani et al., 2017), and it had a service time distribution of Expo(7) (Hamrock et al. 2013). Patients with an ESI level of 5 were simply routed to the “fast track” facility, while all other patients were routed to the main emergency department. This patient process flow, along with the bed respective capacities (listed in Table 2), was utilized in the example model found in Hamrock et al. (2013). So, we decided to design our simulation in an analogous manner. In fact, the number of beds in our simulated facility closely aligned with the available capacities in area hospitals around Savannah, GA (St. Joseph’s/Candler, 2021).

Following triage, patients were routed to the “waiting room” of their assigned facility, which took 5 minutes, to wait for a bed. Once a bed became available, the patients went through the remaining system process. We set up the patient flows in the “fast track” facility and main emergency department in similar ways. Patients were prepped by a nurse in their respective area, they received treatment from a doctor or nurse practitioner, and then an area nurse helped them with consultation and discharge before they left the system. In addition, the capacities of nurses, doctors, and nurse practitioners in both the “fast track” and main emergency departments were directly proportional to the number of beds in each respective area. Once again, the process flow and service times we utilized mirrored the ones described by Dehghani et al. (2017) and Hajirizi et al. (2019). Details for all the service times can be found in Table 4, and the exact capacities for nurses, doctors, and nurse practitioners can be found in Tables 2 and 3. The three major differences between the facilities were: 1. Patients in the main emergency department were treated by a doctor, while those in “fast track” were treated by a nurse practitioner; 2. “Fast track” had one nurse and nurse practitioner assigned to four beds, and the main emergency department had one nurse assigned to three beds and one doctor assigned to four beds; 3. Patients in the main emergency department had the potential to be transferred to imaging or lab testing, and then they had to see a doctor again to discuss their results before going through consultation and discharge.

As mentioned directly above, patients in the main emergency department could be redirected to additional blocks according to specific probabilities. 44% of patients received some time of imaging, 38% of patients received some type of laboratory test, and the remaining 18% of patients simply went straight to consultation and discharge. These percentages were calculated based on figures recorded in Pitts et al. (2018). To simplify the model, we assumed that patients would only go through one route exactly one time (e.g., patients assigned to imaging exclusively took this route and did so just once). If a patient needed lab work, then a “Seize-Delay-Release Process” was done on an emergency department nurse. If a patient needed imaging, then an emergency department nurse was “Seized”, transport to the imaging department took 5 minutes, the patient was directed to a specific machine according to specific probabilities, a “Seize-Delay-Release Process” was done on the selected machine, transport back to the emergency department took 5 minutes, and then the nurse was released. For those patients who were directed to imaging, 56% of these entities needed an Xray, 30% needed a CT scan, 6% needed an ultrasound, 5% needed an MRI scan, and the remaining 3% needed “Other.” These percentages were calculated based on figures recorded in Pitts et al. (2018) and Raja et

al. (2014). The process flow for lab tests and imaging was like the one used by Hajirizi et al. (2019), and the appropriate service times and capacities were recommended by Dehghani et al. (2017) (further details can be found in Table 4). Finally, all patients that needed imaging or a laboratory test were required to wait one hour for their results and speak to a doctor a second time before going through consultation and discharge.

Up to this point, the discussion in this section has focused on all the similarities between the base and pandemic models. However, we made several major adjustments to the pandemic model to account for factors a hospital might have implemented due to the consequences caused by the COVID-19 pandemic. The first set of adjustments was related to the addition of COVID testing at the triage station. We assumed that during triage, the nurses would administer rapid COVID tests to the patients. With the added time of administering the test and waiting for results, we adjusted the service time at the triage station to Expo(12). Next, if patients tested posted for COVID, then they would be assigned to the COVID quarantine area of either the "fast track" facility or main emergency room. Instead of increasing the capacities of the beds, the number of beds in both departments were simply redirected to establish the quarantine areas. We made this adjustment to better represent the resource constraints occurring in the real world. For example, the main emergency department in both systems had a total of 24 beds, yet 6 of the 24 beds in the pandemic model were placed in the COVID quarantine area. Please refer to Tables 2 and 3 for further details about how capacities were redirected in the pandemic system. Also, we assumed that patients tested positive for COVID with a probability of 8.5%. Katella (2021) provided a thorough description of the vaccines currently available in the U.S., and the 8.5% figure was calculated based on an average of the reported efficacies of these three vaccines. Lastly, to represent the chance that our hospital resources, e.g. doctors and nurses, might contract or test positive for COVID while treating patients, we used the concept of "fake customers" that would randomly come in during the simulation to impact our resources. If a doctor or nurse was tagged with COVID by a fake customer, then there would be a nominal impact to the system for a shift of 12 hours. We assumed that in order to fill an empty resource due to sickness, an alternate resource would be identified and in place after 12 hours. To review visualizations of the simulation process flows, please refer to Appendix B.

Output Data Collection & Analysis

** Note that means and half widths are reported throughout with a 95% Confidence Level.*

In simulating a generic hospital emergency room model(Figure 1), we were able to analyze the impact of resource capacity, and patient statistics for a base model, and then compare it to an alternative model that simulated changes a hospital might have implemented or experienced due to pandemic related adjustments. Ideally, as a hospital was planning to add additional resources, they would want to identify potential bottlenecks to plan around them to keep patient wait times reasonable and avoid the need to turn patients away for care. To determine whether differences between queue waiting times and time to see a resource were significantly different between our systems we leveraged a simulation design that utilized common random numbers for generation of both inter-arrival rate and internal process time distributions (Figure2). Since the systems were run dependently, we calculated the difference in means for targeted KPIs at a 95% confidence level (Tables 6 and 9).

What we found during our model analysis is that the Triage process (Figure 4) was a significant bottleneck after adding time for the COVID testing. Since the difference of means

interval for the TriageProcess.Queue, [-4279.0634, -3413.7094] (Table 6), does not contain zero, and is negative, we can conclude that the difference of means is significant and that the pandemic model has the larger average triage queue waiting time. This makes sense since we added time to this process step, and no additional resources. The average wait time for the Triage process queue in the generic pre-pandemic model was 6.2686 +/- 0.21 minutes (Table 5). After making the changes for COVID, the Triage queue time increased to an average of 3852.65 +/- 432.96 minutes! That is over a 60,000% increase.

An explanation for this significant increase might be found by breaking the wait down by the severity of the patient. In this model, patients are triaged in order of their severity, not in a First in First Out (FIFO) order. A more critical patient would arrive in the queue and "jump the line". This meant that the less critical patients remained in the queue longer as they would get continuously bumped back due to patient arrivals with the more critical severity index, (i.e. someone having a heart attack would trump someone with a broken appendage, etc.). All patient severities, except for ES5, experienced significant increases in time to see a doctor, or in the ES5 case, a Nurse Practitioner (Table 9). We conclude that there was a significant increase in time for ES1-ES4 patients as, again, the mean difference between the systems time to see doctor/practitioner does not include zero and is negative indicating that the difference is both significant and that the pandemic model has a higher mean time (Table 9) . Due to the prioritization of patients based on severity, the time to see a doctor expectedly increases by that rating (Table 7). The ES4 patients experienced the greatest impact in the mean time to see a doctor. The ES4 mean time to see a doctor (Table 7) increased from 39.1841 +/- 0.92 minutes, to 15947.15 +/- 2120.31 minutes, a wildly larger and unrealistic time for a patient to wait! We propose that realistically, in this situation, the less critical patients just wouldn't be served as they would likely leave the system in favor of an alternative care solution or the waiting room would fill to capacity and patients would get turned away. This could be added to future simulations to capture that metric.

The second difference noticed between the baseline and pandemic models is that we had a significant increase in the wait time for an emergency room bed (Main ED Bed.Queue). The baseline model had a mean Main ED Bed.Queue wait time of 2.2581 +/- 0.23 minutes, whereas the pandemic model indicated a mean wait time of 43.68 +/- 7.65 minutes, over an 1800% increase (Table 5). Again, the difference in system means (Table 6) for the Seize Main Bed.Queue leads us to conclude that there is a statistically significant difference between the models and that the pandemic case has a longer average wait for a bed. It is important to note that in the pandemic case, we still had the same inter-arrival rates, but we assumed that we needed to take the number of total beds available and split them, making some beds only available for COVID cases (Table 3). When doing this, the mean wait for a main ED COVID bed ended up being very low (zero), compared to the main bed (Table 5). Our pandemic model may have over-compensated for the number of COVID positive patients that would need to be seen.

In order to better optimize the pandemic emergency room, we decided to run a what-if scenario. We decided to investigate increasing resources at the triage process, triage stations and nurses, to help address the large bottleneck found at that part of the simulation. We increased the triage stations and nurses from 2 to 4 (Table 3). Doing this reduced the waiting time for the Triage Process queue to from 6.3210 +/- 0.17 minutes to 1.9981 +/- 0.06 minutes as well as reduced the Patient Time in System (Table 8), less time than took in the pre-pandemic state! However, the difference in system means for the triage process indicates that the

difference between the mean triage times is still significant, but in this scenario, the non-pandemic model had the higher triage queue times. We might have over-compensated with the new resources.

Conclusions

While improvements were made with the what-if experiments, the time in the system was still too high, at 1018.28 +/- 224.17 minutes (around 16 hours!) (Table 8). Adding resources to the triage cleared the triage queue bottleneck, but also uncovered additional downstream bottlenecks for beds and reduced, but did not resolve the issue of long, unrealistic wait times to see a doctor.

The results of our simulations indicated that further optimization is required. Due to the further increase in the main ED bed queue, the imbalance of bed queues, the long time for lower priority ES4 patients to see a doctor, and an average time in system that would require turning patients away , areas that we would propose focussing on for further what-if studies would be the following;

1. Investigate another level of resources at the Triage station. Increasing them to the level in the What-If model created a larger bottleneck of patients waiting for beds. The system might tolerate a slightly higher waiting time at the triage process.
2. Balance the number of COVID / NON COVID beds with the localities 14 day running average positive rate (In this case, we might increase the number of MAIN ED Beds and decrease the number of COVID beds
3. Evaluate options for referrals of non-emergent care, where a doctor is required to partner doctor offices vs. Increasing the number of doctors.

Appendix A: Tables

Table 1

Emergency Severity Index (ESI) - 5 Level Triage Rule

LEVEL	PRIORITY	CASE RATE	DESCRIPTION	ADMISSION RATE
ESI 1	1	2%	Severely unstable, must be seen immediately by a physician, often require an intervention (i.e. intubation) to be stabilized.	73%
ESI 2	2	22%	Potentially unstable, must be seen promptly by a physician (within 10 minutes), often require laboratory and radiology testing, medication, and (often) admission.	54%
ESI 3	3	39%	Stable and should be seen urgently by a physician (within 30 minutes), often require laboratory and radiology testing, medication, and are most often discharged.	24%
ESI 4	4	27%	Stable, may be seen non-urgently by a physician (or MLP), require minimal testing or a procedure, and are expected to be discharged.	2%
ESI 5	5	10%	Stable, may be seen non-urgently by a physician (or MLP), require no testing or a procedure, and are expected to be discharged.	0%

Note: Reiter & Scaletta, 2019

Table 2

Simulation Resources (Generic Emergency Room Model)

Resource Name	Type	Capacity
Admin Staff	Fixed Capacity	2
Triage Station	Fixed Capacity	2
Triage Nurse	Fixed Capacity	2
Fast Track Nurse	Fixed Capacity	2
Fast Track Bed	Fixed Capacity	8
Main ED Bed	Fixed Capacity	24
Main ED Nurse	Fixed Capacity	8
Doctor	Fixed Capacity	6
Nurse Practitioner	Fixed Capacity	2
XRAY, CT, MRI, US, Other	Fixed Capacity	2

Table 3

Simulation Resources (Pandemic Emergency Room Model)

Resource Name	Type	Capacity / Initial	Capacity / What-If
Admin Station_P	Fixed Capacity	2	
Admin Staff_P	Fixed Capacity	2	
Triage Station_P	Fixed Capacity	2	4
Triage Nurse_P	Fixed Capacity	2	4
Fast Track Nurse_P	Fixed Capacity	2	
Fast Track Bed_P	Fixed Capacity	6	
Fast Track COVID Bed_P	Fixed Capacity	2	
Main ED Bed_P	Fixed Capacity	18	
Main ED COVID Bed_P	Fixed Capacity	6	
Main ED Nurse_P	Fixed Capacity	8	
Doctor_P	Fixed Capacity	6	
Nurse Practitioner_P	Fixed Capacity	2	
XRAY, COVID Testing Station	Fixed Capacity	2	
CT, MRI, US, Other	Fixed Capacity	2	

Table 4

Simulation Global Attributes / Distributions

Attribute Name	Distribution (Minutes)
AdminProcess	LOGN(5, 2)
TriageProcess (No Covid Testing)	EXPO(7)
TriageProcess (Covid Testing ~ 5 Minutes Addtl)	EXPO(12)
PatientPrep	EXPO(12)
DoctorTreatment and NursePractTreatment	TRIA(5, 9, 12)
Labtests	TRIA(3,5,7)
ImagingProcess	EXPO(12)
Consultation and Discharge	TRIA(4,5,10)
COVID Rates	8.5% Positive (True)

Table 5

Generic Emergency Rom Vs. Pandemic Emergency Room Models - Queues

Waiting Times In Queue							
Baseline Queues - Pre-Pandemic	**Average**	**Half Width**	**Variance**	**Min Avg**	**Max Avg**	**Min Value**	**Max Value**
TriageProcess.Queue	6.2686	0.2100	0.0898	5.8216	6.9012	0.0000	540.8200
Seize Main ED Bed.Queue	2.2581	0.2300	0.1026	1.8365	2.7432	0.0000	423.2400
Seize Fast Track Bed.Queue	0.0199	0.0400	0.0026	0.0000	0.1649	0.0000	74.6032
Pandemic Queues							
TriageProcess_P.Queue	3852.6500	432.9600	366366.3978	2807.6000	4709.8500	0.0001	39289.3500
Seize Main ED Bed_P.Queue	43.6800	7.6500	114.3130	33.2569	66.8171	0.0000	1598.5900
Seize Main ED COVID Bed_P.Queue	0.0913	0.0600	0.0070	0.0062	0.2744	0.0000	84.8134
Seize Fast Track Bed_P.Queue	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Seize Fast Track COVID Bed_P.Queue	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Baseline Queues - Pre-Pandemic What If	**Average**	**Half Width**	**Variance**	**Min Avg**	**Max Avg**	**Min Value**	**Max Value**
TriageProcess.Queue	6.3210	0.1700	0.0567	6.0169	6.7094	0.0000	448.6800
Seize Main ED Bed.Queue	2.1707	0.2200	0.0947	1.7080	2.6038	0.0000	253.4600
Seize Fast Track Bed.Queue	0.0152	0.0100	0.0012	0.0000	0.1124	0.0000	44.3951
Pandemic Queues - What If Scenario							
TriageProcess_P - WhatIf.Queue	1.9981	0.0600	0.0070	1.8492	2.1332	0.0000	239.4900
Seize Main ED Bed_P- WhatIf.Queue	1103.9800	270.3200	142812.8228	560.0100	1970.8700	0.0000	10390.6600
Seize Main ED COVID Bed_P- WhatIf.Queue	0.1852	0.0500	0.0049	0.0641	0.2769	0.0000	95.3737
Seize Fast Track Bed_P- WhatIf.Queue	0.0082	0.0100	0.0003	0.0000	0.0604	0.0000	40.3934
Seize Fast Track COVID Bed_P- WhatIf.Queue	0.0057	0.0100	0.0002	0.0000	0.0382	0.0000	6.2587

Note: Sourced from output from Arena simulations and additional output analysis

Table 6

Difference in Mean Queue Waiting Times - Systems with Common Random Number

System - Difference of Means			
(time in minutes)	**Mean Difference**	**Mean Variance**	**Half Width**
Triage Differences - Original	-3846.3864	366206.8845	432.6770
Main ED Bed Differences - Original	-41.4219	112.5307	7.5847
Fast Track Bed Differences - Original	0.0076	0.0003	0.0122
	Mean Difference	**Mean Variance**	**Half Width**
Triage Differences - What If	4.3229	0.0529	0.1645
Main ED Bed Differences - What If	-1101.8123	142798.3909	270.1856
Fast Track Bed Differences - What If	0.0041	0.0016	0.0282

Note: Calculated in Queues.xlsx

Table 7

Generic Emergency Room Vs. Pandemic Emergency Room Model - KPIs

Interval	Average	Half Width	Minimum Average	Maximum Average	Minimum Value	Maximum Value
ES1 Time To Doctor	26.9817	0.45	26.0055	28.1245	5.5802	121.82
ES1 Time To Doctor_P	38.7605	0.71	37.1674	40.8960	7.0637	146.17
ES2 Time To Doctor	27.5657	0.23	27.1247	28.2243	5.0974	161.98
ES2 Time To Doctor_P	42.0377	0.11	41.8207	42.2554	5.8555	194.79
ES3 Time To Doctor	29.9670	0.25	29.3139	30.3047	5.0802	164.16
ES3 Time To Doctor_P	74.4377	1.57	70.4191	78.2097	5.4537	421.93
ES4 Time To Doctor	39.1841	0.92	37.3558	41.8994	5.0374	455.66
ES4 Time To Doctor_P	15947.15	2,120.31	11024.10	20324.54	18.9284	39327.58
ES5 Time To Nurse Practitioner	41.6610	1.36	39.2710	44.8606	5.0695	566.84
ES5 Time To Nurse Practitioner_P	84.3656	190.83	0.00	843.66	0.00	1396.15
Pandemic Patient Time in System	3999.14	429.39	2969.37	4846.17	17.9723	39427.42
Pre Pandemic Patient Time in System	114.47	0.65	112.62	115.93	15.7605	581.47

Table 8

Generic Emergency Room Vs. Pandemic What-If Emergency Room Model - KPIs

Interval	Average	Half Width	Minimum Average	Maximum Average	Minimum Value	Maximum Value
ES1 Time To Doctor	26.9304	0.55	25.9067	27.8869	5.2844	106.78
ES1 Time To Doctor_P	36.1947	0.47	35.1915	37.1793	5.9563	163.63
ES2 Time To Doctor	27.5390	0.13	27.3447	27.9230	5.0752	171.84
ES2 Time To Doctor_P	37.1914	0.22	36.6591	37.6881	5.2542	181.65
ES3 Time To Doctor	29.9716	0.21	29.5308	30.4517	5.0373	187.97
ES3 Time To Doctor_P	50.0750	0.69	49.0156	51.6987	5.2540	316.22
ES4 Time To Doctor	38.8666	0.80	37.3584	40.6012	5.0416	323.65
ES4 Time To Doctor_P	3480.10	889.28	1722.40	6349.15	5.4048	10423.09
ES5 Time To Nurse Practitioner	42.1883	0.59	41.4286	44.3504	5.0666	463.24
ES5 Time To Nurse Practitioner_P	33.7304	0.30	33.0562	34.3302	5.1228	307.12
Pandemic Patient Time in System	1018.28	224.17	570.59	1741.91	16.5813	10522.09
Pre Pandemic Patient Time in System	114.45	0.33	113.93	115.24	16.4444	479.27

Table 9

Difference in Mean Times - Systems with Common Random Numbers

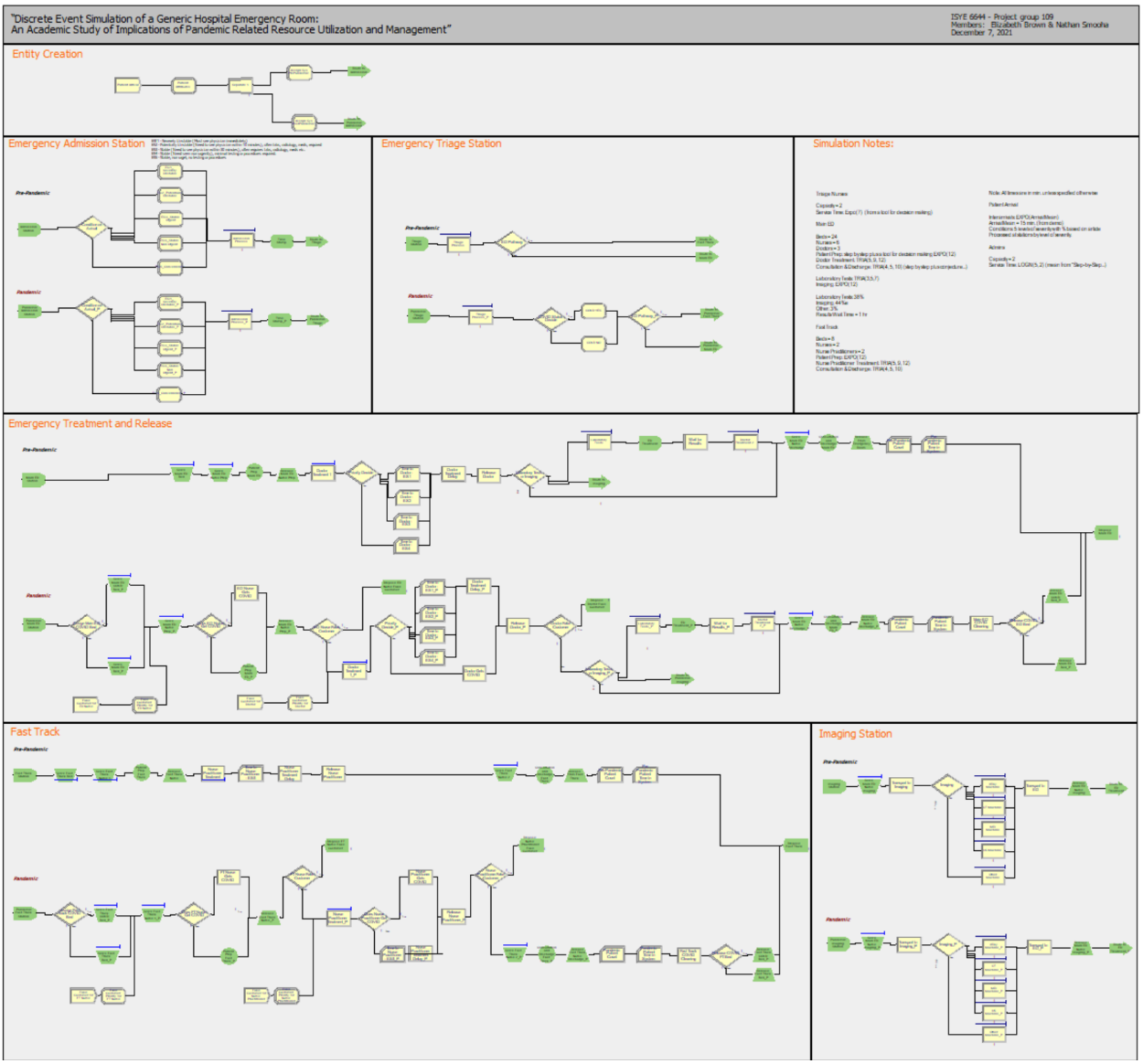
System - Difference of Means			
(time in minutes)	**Mean Difference**	**Mean Variance**	**Half Width**
ES1 Time To Doctor - Original	-11.7788	1.1918	0.7805
ES2 Time to Doctor - Original	-14.4720	0.1183	0.2459
ES3 Time to Doctor - Original	-44.4707	4.1516	1.4568
ES4 Time to Doctor - Original	-15907.9619	8782875.8815	2118.9413
ES5 Time	-42.7049	71627.9364	191.3559
Patient Time in System - Original	3884.7650	360483.5211	429.2826
	Mean Difference	**Mean Variance**	**Half Width**
ES1 Time To Doctor - What If	-9.2643	0.8684	0.6663
ES2 Time to Doctor - What If	15.3158	6239.6135	56.4780
ES3 Time to Doctor - What If	-20.1036	0.9247	0.6876
ES4 Time to Doctor - What If	-3441.2364	1545818.1421	888.9550
ES5 Time	8.4579	0.8947	0.6763
Patient Time in System - What If	903.8300	98250.3511	224.1133

Note: Calculated in Queues.

Appendix B: Figures

Figure 1

Simulation Overview



Note: Project_Generic_Emergency_20211116_baseline.doe

Figure 2

Simulation Entity Creation - Common Random Numbers

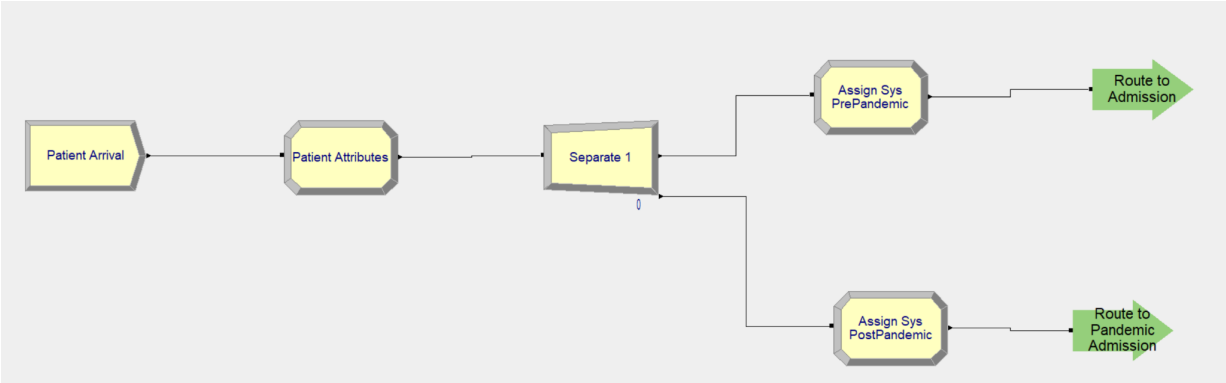


Figure 3

Simulation Emergency Admission Station - Severity Index Assignment

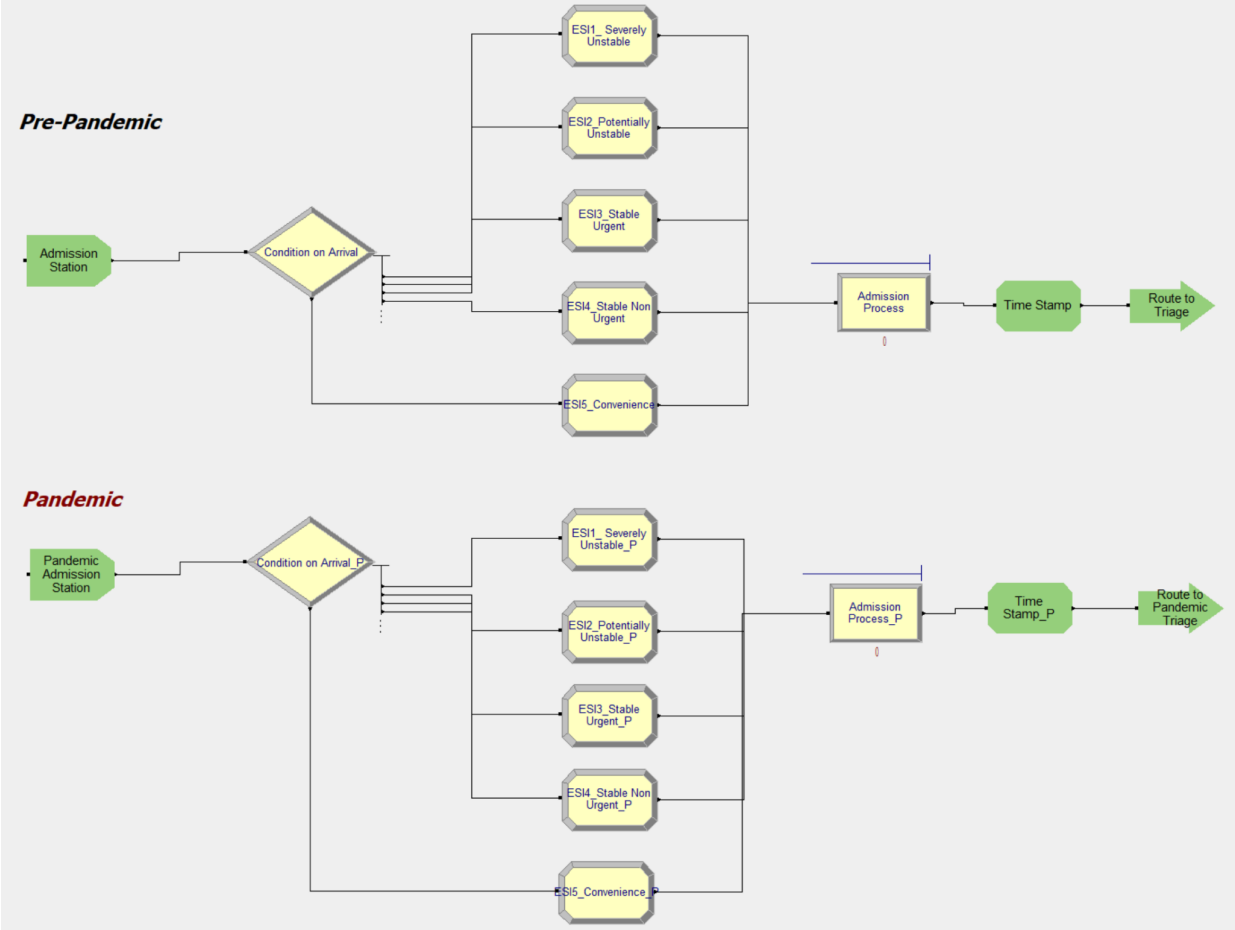


Figure 4

Simulation Emergency Triage Station with Pandemic COVID Testing

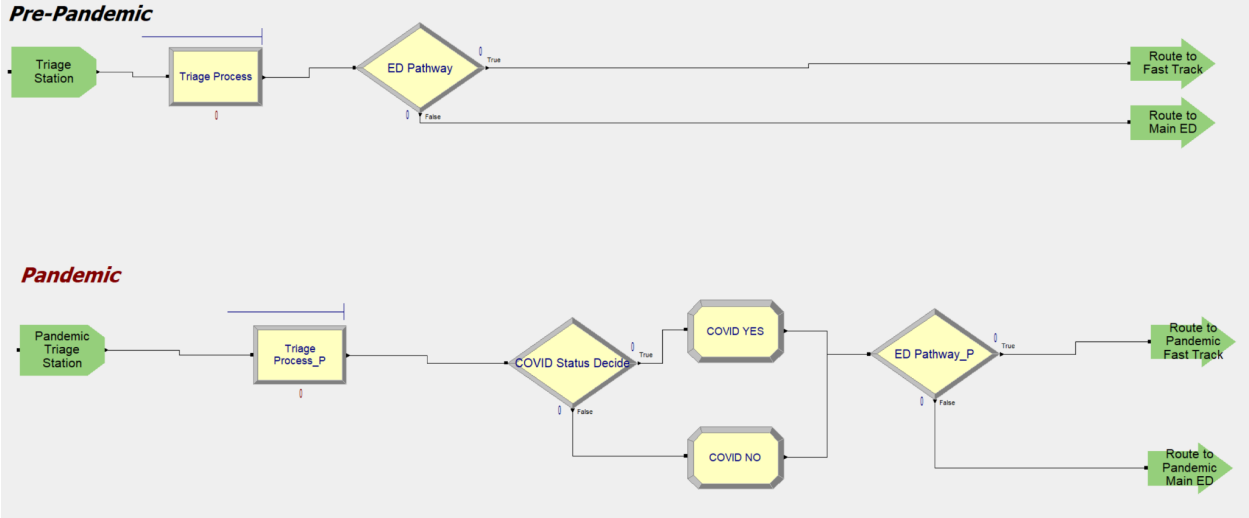


Figure 5

Simulation Emergency Treatment and Release

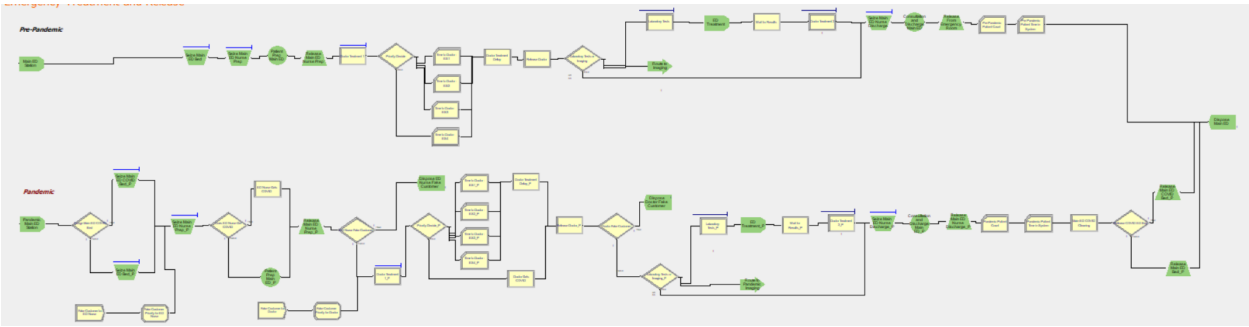


Figure 6

Simulation Emergency Imaging

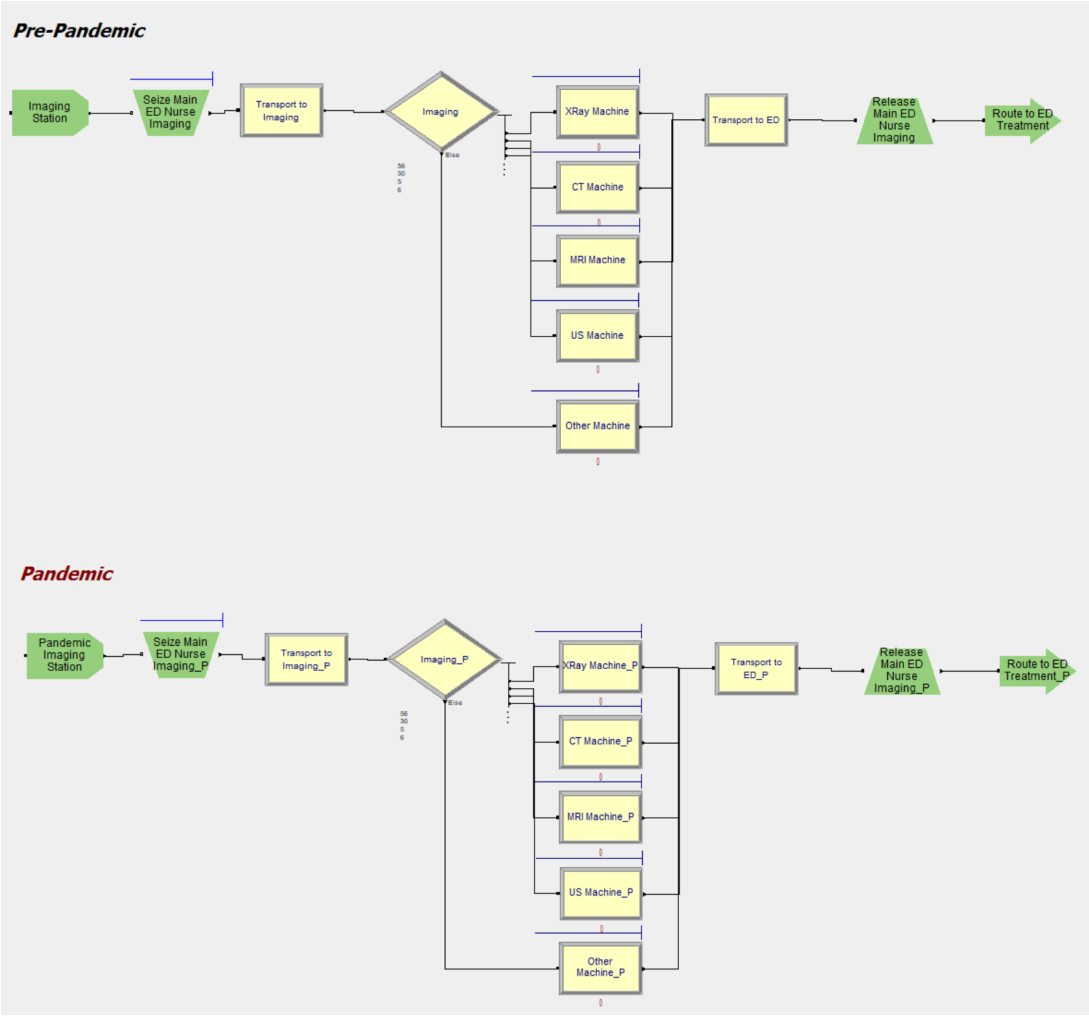
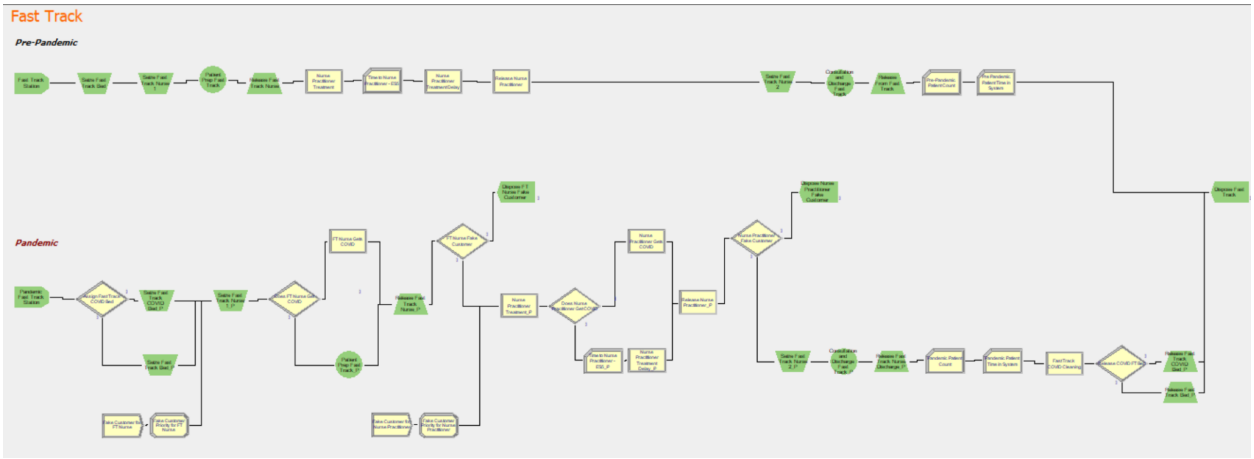


Figure 7

Simulation Emergency Fast Track



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